

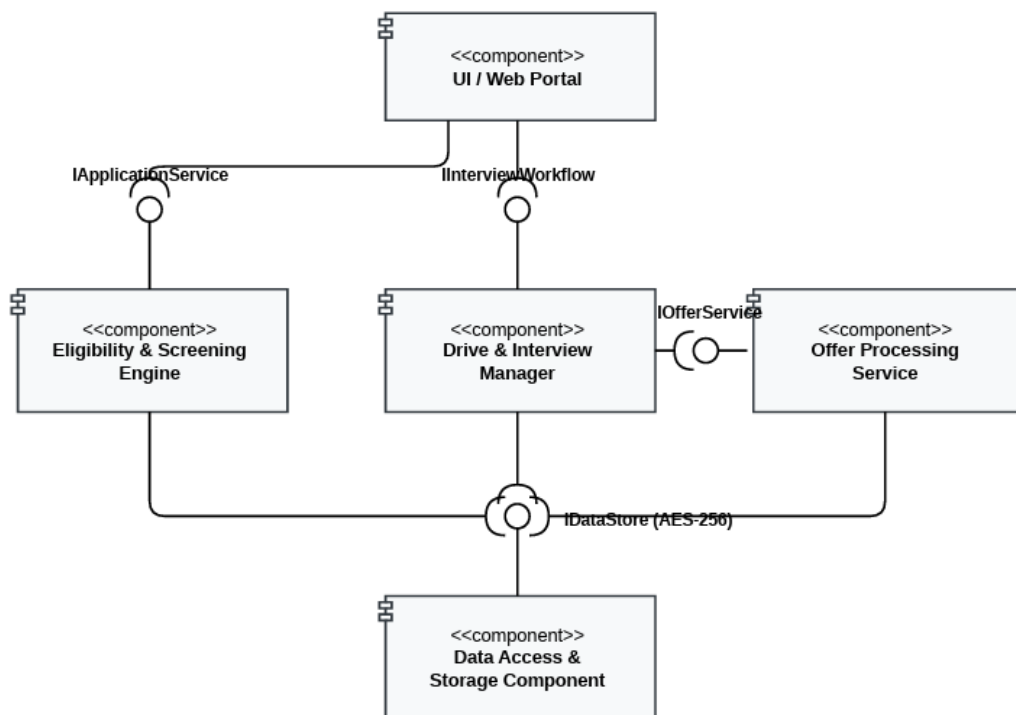
Lab 3 - UML Component Diagram

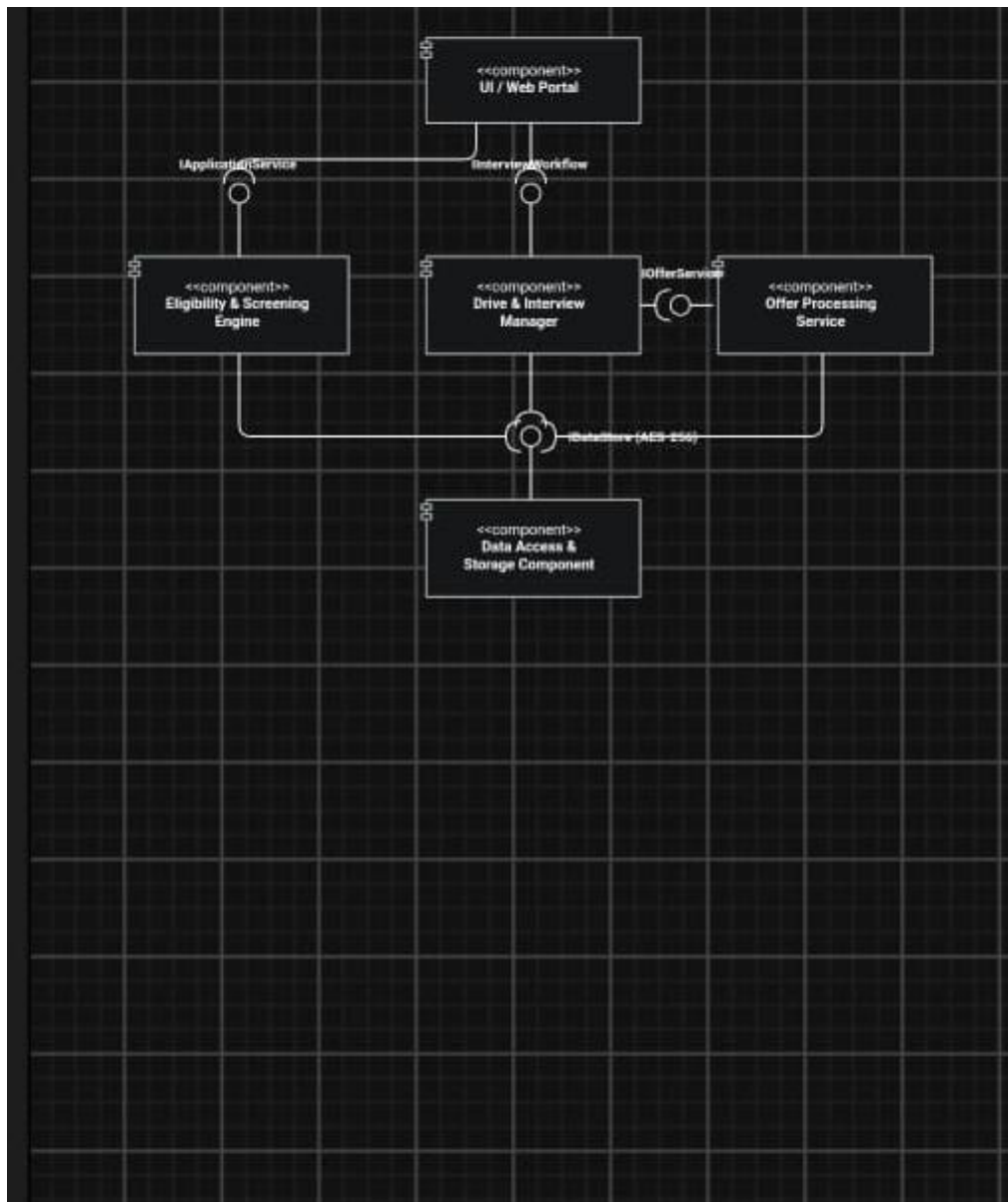
(Campus Placement & Internship Pipeline)

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UML COMPONENT DIAGRAM





Justification:

Architecture Selection: I chose a Layered Architecture for the Campus Placement & Internship Pipeline.

- **Architectural Choice:** Layered (3-Tier) Architecture.
- **Scenario-Specific Reasons:**
 1. *Separation of Academic Logic:* Decouples eligibility rules (CGPA cutoffs, backlog verification) from UI and database layers, allowing rule modifications without altering user flows.
 2. *Role-Based Access Isolation:* Clearly separates student self-service features (resume upload, drive applications) from administrative controls (interview pipeline transitions, offer generation).

- **Security Advantage:** Fulfills NFR-001 by centralizing database transactions inside a dedicated data access layer enforcing AES-256 encryption at rest for student profiles and sensitive offer letters.
- **Performance Benefit:** Directly addresses NFR-002 by enabling intermediate business components to cache active drive eligibility rules, comfortably sustaining 500 concurrent student applications within the 3-second latency limit

Component & Interface Mapping Table

Component DOCX+ 1	Provided Interface (—○) DOCX	Required Interface (-(-) DOCX	Traceability PDF
UI / Web Portal	—	IApplicationService, IInterviewWorkflow	Student/Admin UI
Eligibility & Screening Engine	IApplicationService	IDataStore	FR-001, FR-002
Drive & Interview Manager	IInterviewWorkflow	IOfferService, IDataStore	FR-004
Offer Processing Service	IOfferService	IDataStore	FR-005
Data Access & Storage	IDataStore (AES- 256)	—	NFR-001